Zr3i

CROP AND FARM RISK INDEX

Methodology Specification

From observation to index — definitions, measurement, weighting, worked examples, and interpretation for lenders and insurers.

Method version	FRI-2026-01
Document status	For model validation review
Prepared for	Credit risk and model validation teams; actuarial and underwriting teams
Issuer	DigiWell for Digital Solutions LLC (trading as Zr3i), CR 173893, Cairo
Date	27 September 2026

> *This document describes the specified model. Where the running implementation diverges from this specification, the divergence is recorded in Annex A with file-level evidence. Annex A should be read as part of the specification, not as an appendix to it.*

Contents

1. Purpose, scope and status

The Crop Risk Index (CRI) and the Farm Risk Index (FRI) are production-risk measures. They estimate the likelihood that a parcel, or a farm, produces materially less than its own recent norm, using observable growing conditions. They are designed to be consumed by institutions that already carry agricultural exposure and need an evidence base for it.

1.1 What this document covers

- Every input the model consumes, how it is measured, and at what cadence.
- Every intermediate feature, with its formula, units and admissible range.
- The seven components, what each represents, and the evidence basis for its weight.
- The arithmetic of the CRI and the FRI, with fully worked examples.
- How a lender and an insurer should — and should not — use the output.
- The calibration and validation protocol, and the governance around method versions.

1.2 Status of the model

The specification below is complete and versioned. The implementation is built and running. The empirical calibration against harvest outcomes has not yet been performed: the per-crop error terms carried in the specification are design targets, not fitted values. Any statement in this document of the form "calibrated against yield shortfall" should be read as "specified to be calibrated against yield shortfall, with calibration scheduled for the 2026/27 season".

> *A validation team should treat the current index as a specified but uncalibrated model. It is internally consistent and reproducible; it is not yet empirically validated. Section 13 sets out the protocol and the single acceptance test that determines whether it becomes so.*

2. Definitions

2.1 Crop Risk Index (CRI)

Attribute	Definition
Unit of observation	One parcel, one crop, one season.
What it represents	The likelihood that this parcel under-produces relative to its own recent norm, given observable growing conditions.
Scale	0 to 100, expressed as the parcel's percentile against a multi-season reference distribution for its crop and governorate.
Direction	Higher means more risk. 80 means this parcel ranks worse than 80% of comparable parcel-seasons. This is not a credit score.
Construction	Weighted mean of seven component scores, renormalised over the components that could be established, then expressed as a percentile against the reference distribution (section 10).
Validated against	Realised yield shortfall — production below 80% of the parcel's own five-season median.

2.2 Farm Risk Index (FRI)

Attribute	Definition
Unit of observation	One farm, one season.

Attribute	Definition
What it represents	The likelihood that the farm as a whole under-produces, including structural risks no single parcel reveals.
Scale and direction	0 to 100 percentile against a farm-level reference distribution; higher means more risk.
Construction	Area-weighted mean of the farm's parcel CRIs, plus farm-level adjustments.
Exclusions	Parcels whose season data fails the validity floor are excluded from the mean, not estimated.

2.3 Why two indices

A parcel can be individually sound while the farm is fragile. Nine healthy parcels drawing from a single canal command is one point of failure, and no parcel score can see it. Conversely, a farm can be sound in aggregate while one block carries most of the downside. The CRI is blind to the first; a farm-only measure would hide the second.

2.4 What neither index is

Neither index is a probability of default, a collateral value, an approval, a limit, a rate, a credit score, a yield forecast, or an insurance premium. The specification file carries this restriction in its own disclaimer field. The reason is structural rather than cautionary: the model is estimated against agronomic outcomes, not against lending outcomes, so it cannot carry information it was never fitted to.

3. Model architecture

The model has four layers. Each layer is separately versioned and separately auditable, and every value carries the identifier of the method version that produced it.

Layer	Contains	Example
0. Verification	Field observations used to key, qualify and calibrate everything above. Held out of the scoring chain by design (section 4).	A ground-truth sample point; an accuracy register entry.
1. Observation	Raw measurements from satellite, sensors, field visits and registers.	A Sentinel-2 surface-reflectance scene; a harvest record; a parcel boundary.
2. Feature	Season-level quantities derived from observations.	ndmi_min; yield_shortfall_frequency; area_variance_pct.
3. Component	Seven interpretable risk dimensions, each scored 0–100.	Water security; Climate exposure; Tenure and rights.
4. Index	CRI per parcel-crop-season; FRI per farm-season; band A–E.	CRI 50.3, band C.

The separation matters for validation. A change to a scoring ramp is a change at layer 3 and does not disturb layers 1 and 2. A change to how a feature is computed invalidates every index value derived from it, and the method version must increment.

4. Ground truth and field verification

Ground truth is field observation of what is actually growing, recorded at a known location and time. It has four distinct roles in this model, and they are separated deliberately because they carry different methodological obligations. Conflating them is the most common error in satellite-derived agricultural risk products.

Role	What it does	Why it is separated
1. Crop identity	Confirms which crop is actually in the ground, against what was declared.	Crop selects the heat threshold, the phenological window, the price book and the per-crop error term. A keying input, not a fitting input.
2. Accuracy quantification	Measures agreement between the model and the field, per crop, governorate and size stratum.	Held out of the model by design. Data used to fit a model cannot then measure it.
3. Yield calibration	Links the vegetation signal to realised tonnes per feddan.	Converts an ordinal index into something with economic meaning. Performed once per calibration cycle, not per season.
4. Boundary confirmation	Verifies that the parcel is where, and the size, it is claimed to be.	Supports the tenure component and every area-weighted quantity.

4.1 Why ground truth is not an input to the score

Roles 2 and 3 are validation functions. In map-accuracy practice the reference sample is held out of model construction: a sample used to build the classifier cannot then be used to measure it without inflating the measured accuracy. Ground truth therefore does not feed any feature, component or index value, and this is a property of the design rather than an omission.

> *This is a claim a validation team will test. The correct answer is that the accuracy register is computed from a sample drawn at random and walked blind to the model prediction, and that no ground-truth observation enters the feature layer. Walking only the parcels the model has flagged would measure false positives and never false negatives — which is the half a lender cares about.*

4.2 Sampling design

Element	Specification
Selection	Stratified random. Points are drawn, not chosen; field teams do not select which parcels to visit.
Strata	Crop × governorate × parcel size band.
Blinding	The field record is captured without sight of the model prediction for that parcel.
Evidence	Photographs with heading and GPS timestamp, bound to the sample point record.
Window	November to April, aligned to the winter season.
Publication	Agreement is published per stratum, including strata whose agreement is too low to rely on.

4.3 What is not yet built

Two of the four roles are not operational. Crop identity is currently taken from the parcel profile as a declared value and is never reconciled against a field observation — recorded as Gap A-1. Yield calibration has not been performed, which is why no statement in this document converts an index value into an expected yield or an expected loss. The accuracy register exists and functions; boundary confirmation is partially served by the measured-versus-declared area comparison in section 6.

4.4 Confidence, and where it belongs

Confidence is read from the published accuracy register for the parcel’s stratum. Where no register entry exists the system returns "not yet established" rather than a value — confidence is never assigned, inferred or defaulted. At present confidence is surfaced as a separate attribute rather than travelling with each index value; this is recorded as Gap A-9.

5. Layer 1 — Observation inputs

Every input below is defined by what it physically measures, how it is obtained, and its cadence and resolution. Inputs marked as register data are declared by a party and are treated as assertions to be verified, not as measurements.

5.1 Satellite observation

Input	What it measures	Source and method	Cadence / resolution
Surface reflectance	Fraction of incident solar radiation reflected by the surface, per spectral band.	Sentinel-2 MSI Level-2A, atmospherically corrected. Scenes are masked using the Scene Classification Layer; cloud, shadow, snow, saturated and no-data classes are excluded, and the mask is dilated one pixel at 20 m.	5 days at the equator; 10–20 m depending on band
Parcel statistic	A single value per parcel per pass, not a per-pixel value.	Zonal median over the parcel polygon after an inward buffer of 15 m to remove edge contamination. Median is used rather than mean so a few surviving cloud pixels cannot move the value.	Per valid pass
Valid fraction	Share of the parcel's pixels that survived masking on a given pass.	Count of retained pixels divided by total pixels within the buffered polygon.	Per pass; dimensionless 0–1

> *The valid fraction is a data-quality measure. It qualifies every optical value computed from that pass and is used to decide whether a parcel-season may be scored at all. It is not itself a risk signal and must never be scored as one.*

5.2 Spectral indices used

Index	Formula	What it indicates
NDVI	(B8 − B4) / (B8 + B4)	Canopy greenness and photosynthetic capacity. Proxy for biomass accumulation.
NDMI	(B8A − B11) / (B8A + B11)	Canopy water content. Responds to plant water status before visible wilting.

Both are published formulations and neither is proprietary to Zr3i. The proprietary element is the masking, aggregation and version-binding discipline applied to them, not the arithmetic.

5.3 Climate observation

Input	What it measures	Source and method
Heat stress indicator	Exposure to temperatures above the crop's damage threshold during its heat-sensitive phenological window.	Daily maximum air temperature from a gridded reanalysis product, intersected with the parcel and the modelled phenological stage. Normalised to 0–1 against the crop-specific threshold.
Rainfall anomaly	Deviation of season rainfall from the local long-run norm.	Gridded precipitation, accumulated over the season and expressed as a standardised anomaly.

5.4 Ground and register inputs

Input	What it measures	Source and method
Harvest record	Realised yield for a parcel in a closed	Farm records, collected at onboarding and each season.

Input	What it measures	Source and method
	season, in tonnes per feddan.	Records flagged in-season or inactive are excluded from all statistics.
Parcel boundary	The physical extent of the parcel.	Drawn or imported polygon in EPSG:4326, projected to the local UTM zone for area computation.
Measured area	Area of the boundary polygon, in feddan.	ST_Area of the projected geometry divided by 4,200.83 m² per feddan.
Declared area	Area asserted by the holder or in the land record.	Register or farmer declaration. Treated as an assertion, not a measurement.
Ground truth point	Crop identity and condition at a known location and time.	Field visit at a randomly drawn sample point, recorded blind to the model prediction, with photographs, heading and GPS time.

> *The distinction between measured and declared area is load-bearing. Their divergence is itself a feature (section 6) and is the only documentary-integrity signal currently in the model.*

6. Layer 2 — Features

A feature is a season-level scalar derived from observations. Features are persisted per parcel, per season, per method version, so any index value can be recomputed and reconciled. Where the inputs for a feature are absent or fail the quality floor, the feature is left null — it is never imputed.

Feature	Definition	Formula / derivation	Units
ndvi_integral	Central tendency of canopy greenness across the season.	Median of the parcel's valid NDVI observations over the season window, multiplied by 100.	0–100 index units
ndmi_min	Canopy water status at its worst point in the season.	Median of the parcel's valid NDMI observations, multiplied by 100. Lower values indicate drier canopy.	0–100 index units
heat_stress	Severity of heat exposure in the sensitive window.	Crop-specific heat indicator normalised to the 0–1 interval, where 1 is the maximum modelled exposure.	0–1 dimensionless
rain_deficit	Shortfall of season rainfall against the local norm.	Standardised rainfall anomaly, normalised to 0–1.	0–1 dimensionless
area_variance_pct	Divergence between measured and declared area.	|area_measured − area_declared| / area_declared × 100, capped at 100.	percent
ndvi_cohort_percentile	Where this parcel sits against its peers this season.	Share of cohort parcels whose season NDVI is at or below this parcel's, expressed as a percentage.	percentile 0–100
yield_shortfall_frequency	How often this parcel has fallen materially short.	Count of closed seasons with yield below 80% of the parcel's own median, divided by the number of closed seasons.	0–1 proportion
worst_shortfall_ratio	How deep the worst shortfall was.	1 − (minimum season yield / median yield), clamped to 0–1.	0–1 proportion
valid_fraction	Usable share of the parcel's observations.	Retained pixels divided by total pixels. Used as a gate, not as a score.	0–1 proportion

6.1 The validity gate

A parcel-season with a valid fraction below 0.60 is marked insufficient. An insufficient parcel is still scored where its components can be established, but it is excluded from the farm roll-up entirely rather than contributing a low-confidence value to a farm average.

$$\text{valid_fraction} \geq 0.60 \quad \rightarrow \quad \text{parcel enters the farm roll-up}$$

$$\text{valid_fraction} < 0.60 \quad \rightarrow \quad \text{parcel excluded from AWCRI}$$

6.2 Minimum history

The two yield features require at least two closed seasons. With fewer, both are null and the yield stability component cannot be established — the CRI is then computed over the remaining six components and renormalised. This is stated explicitly because it changes the effective weighting of every other component, and a validation team must be able to detect when it has happened.

7. Layer 3 — The seven components and their weights

Each component converts one or more features into a 0–100 risk score using a piecewise-linear ramp. A ramp is defined by the feature value at which risk is lowest, the value at which it is highest, and the scores at those two points; values outside the range are clamped. Ramps are deliberately simple and monotonic so that any score can be explained to a credit committee in one sentence.

$$\text{score} = s_lo + \text{clamp}((v - v_lo) / (v_hi - v_lo),\ 0,\ 1) \times (s_hi - s_lo)$$

7.1 Status of the weights

The seven weights are expert-set. They were chosen from the agronomic and remote-sensing literature summarised below, and they are not derived from Zr3i data. They will be re-estimated empirically after the first calibration season, at which point the method version will increment and the prior weights will remain retrievable. Nothing in this document should be read as claiming the current weights were fitted.

> *For a validation team: expert-set weights are defensible for a first-generation model provided the rationale is documented and the re-estimation is committed to. What is not defensible is presenting expert judgement as an empirical result. Section 13 defines how and when these weights are replaced.*

Water security — weight 25

What it represents	The risk that the crop does not receive sufficient water at the times it matters, whether through canal unreliability, allocation shortfall or abstraction limits.
Feature(s)	ndmi_min — canopy water content at its lowest seasonal point.
Scoring ramp	NDMI 15 → score 85 (driest, highest risk); NDMI 45 → score 25. Linear between, clamped outside.

Basis for the weight: Water is the binding constraint on Egyptian irrigated agriculture, and the weight reflects that rather than a general agronomic ranking. FAO's standard treatment models yield as a direct function of evapotranspiration deficit through the yield response factor Ky, making water the only input with a codified, crop-specific yield-loss coefficient. Egyptian field evidence is consistent: water availability determines not only total wheat yield but whether the crop yields at all, and agriculture consumes over 84% of national water resources, so allocation risk is systemic rather than idiosyncratic. Assigning water the largest single weight follows directly.

References: Steduto et al. (2012), FAO Irrigation and Drainage Paper 66; Agronomy 14(12):3057 (2024), Egypt wheat water-input response.

Climate exposure — weight 20

What it represents	The risk that temperature or rainfall during the crop's sensitive phenological window damages yield formation, independent of water supply.
Feature(s)	heat_stress — normalised exposure above the crop threshold in the sensitive window.
Scoring ramp	heat_stress 0 → score 20; heat_stress 1 → score 75.

Basis for the weight: Climate is weighted second because the literature finds inter-annual, weather-driven variation to be the largest single source of yield variability, and larger than spatial soil variation. Global analysis attributes roughly a third of observed yield variability to climate variation alone; within-field studies find temporal variance generally exceeds spatial variance. The mechanism is stage-specific rather than seasonal: in maize, daytime temperature above about 35 °C is the recognised injury threshold, and kernel number is set in a window of roughly thirty days bracketing silking, so exposure inside that window carries disproportionate yield consequence.

References: Ray et al. (2015), Nature Communications 6:5989; Yield penalty of maize under heat stress, The Crop Journal (2022); kernel formation sensitivity around silking, Field Crops Research (2021).

Soil and land — weight 15

What it represents	The risk carried by the land itself — texture, organic matter, salinity, drainage and position — which constrains attainable yield irrespective of season.
Feature(s)	ndvi_integral, read as an expression of the land's realised productive capacity across the season.
Scoring ramp	ndvi_integral 20 → score 75; 55 → score 35.

Basis for the weight: Soil properties are a substantial but sub-dominant driver. Regression of soil properties on yield in winter wheat explains roughly 45–46% of within-field yield variability, which is material but leaves the majority unexplained; and because soil constraints are largely static, they shift the attainable level rather than the year-to-year variance a lender is exposed to. A weight below water and climate, and equal to yield stability, reflects that balance.

References: Plant and Soil (2023), small-scale soil variability and winter wheat yield formation; Scientific Reports 8:14113 (2018), drivers of within-field yield variability.

Yield stability — weight 15

What it represents	The risk implied by the parcel's own track record — how often and how deeply it has fallen short of its own norm.
Feature(s)	Closed-season harvest records: shortfall frequency, depth of worst shortfall, and coefficient of variation.
Scoring ramp	Composite formula given in section 7.2; range 15 (perfectly stable) to 100.

Basis for the weight: This is the only component estimated directly from outcomes rather than from conditions, which is why it carries weight equal to soil despite using a single data source. Interannual variability normalised by mean yield — the coefficient of variation — is the conventional actuarial measure of production risk, and a parcel's own history is the most direct available evidence of its exposure. It is held at 15 rather than higher because it is backward-looking, requires at least two closed seasons, and is unavailable for new parcels.

References: Ray et al. (2015), Nature Communications 6:5989 (use of CV for interannual yield variability).

Management quality — weight 10

What it represents	The risk arising from husbandry decisions within the operator's control — sowing date, crop establishment, season length and rotation.
Feature(s)	ndvi_integral, read as the observable consequence of establishment and husbandry decisions.
Scoring ramp	ndvi_integral 25 → score 55; 55 → score 25.

Basis for the weight: Management effects are real and measurable but smaller and more reversible than the physical constraints above. Sowing date is the clearest single instance: winter wheat yield declines by roughly 1% per day of delay, with a seven-day delay costing 6–7.7%, and a global meta-analysis confirms sowing-date shifts produce consistent but bounded yield effects. A 10% weight reflects an effect that is consistent, agronomically well-attested, and materially smaller than water or climate.

References: European Journal of Agronomy (2020), yield penalty from delayed sowing of winter wheat; Field Crops Research (2024), global meta-analysis of sowing-date shifts.

Tenure and rights — weight 8

What it represents	The risk that the operator's claim to the land, or the accuracy of what has been declared about it, is weaker than represented.
Feature(s)	area_variance_pct — divergence between measured and declared area.
Scoring ramp	area_variance 0% → score 25; 25% or more → score 70.

Basis for the weight: Tenure is weighted low deliberately, and the evidence supports the restraint. A systematic review of land property-rights interventions found strong evidence for effects on investment but an explicit lack of support for links to productivity, credit access and income; a 2026 meta-analysis found the relationship contextual rather than universal, with effects concentrated where institutions are already strong. Tenure therefore belongs in the model — it is a genuine documentary-integrity signal, and area over-declaration is directly observable — but not at a weight implying a strong productivity channel that the literature does not support.

References: Journal of Development Effectiveness 8(1) (2016), systematic review of land property-rights interventions; Frontiers in Sustainable Food Systems (2026), meta-analysis of tenure security and agricultural performance.

Peer and context — weight 7

What it represents	The risk visible only by comparison — whether this parcel is performing in line with comparable parcels facing the same season.
Feature(s)	ndvi_cohort_percentile within the crop and governorate cohort for that season.
Scoring ramp	percentile 0 → score 75; percentile 100 → score 25.

Basis for the weight: Relative position carries the smallest weight because it is a differencing signal: it removes the common seasonal shock and isolates parcel-specific divergence, which is informative but largely already captured by the condition components. Its value is as a control rather than a driver — it prevents a bad regional season from being read as a bad parcel. The remote-sensing credit literature supports the general link between vegetation health indices and agricultural credit risk, which is the basis for including relative vigour at all.

References: World Development 127 (2020), remotely-sensed vegetation health indices and credit risk in agricultural microfinance.

7.2 The yield stability formula

Yield stability is the one component not produced by a simple ramp. It combines three properties of the parcel's closed-season yield series: how often it fell short, how deep the worst shortfall was, and how variable the series is overall.

$$\text{score} = 15 + f \times 45 + d \times 25 + \min(cv, 1) \times 15$$

Term	Definition
f	Shortfall frequency — seasons below 80% of the parcel median, divided by seasons observed.
d	Depth — 1 − (minimum yield / median yield), clamped to 0–1.
cv	Coefficient of variation of the yield series, capped at 1.
15	Floor. A parcel with a perfect record still carries baseline production risk.

The result is clamped to 0–100. The weights inside the formula place most emphasis on frequency, then depth, then general variability — a parcel that fails often is treated as riskier than one that failed once badly.

8. Layer 4a — Computing the CRI

The CRI is the weighted mean of the component scores, renormalised over the weight that could actually be established. Components that cannot be established are recorded as null with their weight, not silently set to a default.

$$\text{CRI} = \Sigma\ (\text{score}_i \times \text{weight}_i) \,/\, \Sigma\ \text{weight}_i \quad \text{over established components only}$$

If all seven are established the denominator is 100 and the expression reduces to a simple weighted average. If, say, yield stability cannot be established for a new parcel, the denominator becomes 85 and the remaining six components are scaled up proportionally. The number of established components and the established weight are both carried with the score.

8.1 Worked example — parcel ZR-KS-03

A 55.7 feddan maize parcel in Kafr El Sheikh with eight closed seasons of harvest records. All seven components are established.

Component	Feature	Value	Score	Weight	Contribution
Water security	ndmi_min	24.0	67.0	25	16.75
Climate exposure	heat_stress	0.58	51.9	20	10.38
Soil and land	ndvi_integral	46.0	45.3	15	6.79
Yield stability	yield history	—	34.8	15	5.22
Management quality	ndvi_integral	46.0	34.0	10	3.40
Tenure and rights	area_variance_pct	12.0	46.6	8	3.73
Peer and context	ndvi_cohort_percentile	34.0	58.0	7	4.06

Established weight = 100. Weighted sum = 5,033.3.

$$\text{CRI} = 5{,}033.3 \,/\, 100 = 50.3 \quad \rightarrow \text{ band C}$$

8.2 Reading the worked example

- Water security contributes 16.75 of the 50.3 — a third of the score from one component, which is the intended consequence of the 25% weight.
- Yield stability scores 34.8 from a history with two shortfalls in eight seasons, a worst season at 73.5% of median, and a coefficient of variation of 0.13.
- Tenure scores 46.6 because the parcel's measured area diverges 12% from its declared area — a moderate documentary flag, not a legal finding.
- No component is at a floor or ceiling, so every one is carrying information. A component pinned at its ramp limit across an entire portfolio is a diagnostic warning, not a result.

9. Layer 4b — Computing the FRI

The FRI starts from the area-weighted mean of the farm's parcel CRIs, then applies farm-level adjustments for structural risks that no parcel score can express.

$$\text{AWCRI} = \Sigma\ (\text{area}_j \times \text{CRI}_j) \,/\, \Sigma\ \text{area}_j \qquad \text{over parcels passing the validity gate}$$

$$\text{FRI} = \text{AWCRI} + \Sigma\ \text{adjustments}$$

9.1 The farm-level adjustments

Adjustment	What it captures	Direction
Crop concentration	Herfindahl index of crop areas. A farm in one crop carries that crop's whole risk.	Positive when concentrated
Spatial concentration	Number of distinct canal commands. Parcels on one command share one failure mode.	Positive when concentrated
Irrigation diversity	Number of distinct water sources available to the farm.	Positive when single-sourced
Record length	Seasons of harvest history on file. Longer records reduce estimation uncertainty.	Negative — reduces the index
Worst-parcel exposure	Share of farm area sitting in the worst band.	Positive when concentrated in poor parcels

9.2 Worked example — Delta Agro Estate

Nine parcels, 428.0 feddan, four crops, all passing the validity gate.

Parcel	Crop	Area (fd)	CRI	Band	Area × CRI
ZR-KS-01	Wheat	62.4	19.0	A	1,185.6
ZR-KS-02	Wheat	48.1	26.0	B	1,250.6
ZR-KS-03	Maize	55.7	50.3	C	2,801.7
ZR-KS-04	Sugar beet	71.2	33.0	B	2,349.6
ZR-KS-05	Wheat	39.8	22.0	B	875.6
ZR-KS-06	Maize	44.3	58.0	C	2,569.4
ZR-KS-07	Rice	52.6	71.0	D	3,734.6
ZR-KS-08	Sugar beet	33.9	29.0	B	983.1
ZR-KS-09	Wheat	20.0	41.0	C	820.0
Total		428.0			16,570.2

$$\text{AWCRI} = 16{,}570.2 \,/\, 428.0 = 38.7$$

Adjustment	Value
Crop concentration (HHI 0.289, four crops)	+1.2
Spatial concentration (all parcels, one canal command)	+3.1
Single irrigation source	+2.0
Record length, 8 seasons	−5.8

Adjustment	Value
Worst-parcel exposure (52.6 fd, 12.3%, band D)	+1.2
Σ adjustments	+1.7

$$\text{FRI} = 38.7 + 1.7 = 40.4 \quad \rightarrow \text{ band B}$$

9.3 The concentration insight

Excluding ZR-KS-07 — the single rice block, 52.6 feddan, CRI 71 — the area-weighted mean falls to 34.2 and the FRI to 34.7. One parcel representing 12.3% of farm area accounts for 5.7 points of the farm index. This is the kind of statement the FRI exists to produce: it tells a credit officer where the exposure actually sits, rather than only how the farm scores overall.

> *The value 40.4 in this example falls in the gap between the published band cutpoints (B ends at 40, C begins at 41). The implementation returns band A for this value, which is a defect. It is recorded as Gap A-2 in Annex A. The band shown above is the intended band under the specification.*

10. Scale, distribution and bands

This section sets out the distributional properties of the index and the banding rule. It is placed before the institutional sections deliberately: a rating scale is only useful to the extent that it separates the population, and the separation is a property of the construction rather than of the data.

10.1 The working score is bounded and compressed

Every component is a clamped linear ramp with a hard floor and ceiling. Combining them with the published weights therefore produces a working score that cannot reach the ends of its own nominal scale.

Component	Weight	Score floor	Score ceiling	Weighted floor	Weighted ceiling
Water security	25	25.0	85.0	6.25	21.25
Climate exposure	20	20.0	75.0	4.00	15.00
Soil and land	15	35.0	75.0	5.25	11.25
Yield stability	15	15.0	100.0	2.25	15.00
Management quality	10	25.0	55.0	2.50	5.50
Tenure and rights	8	25.0	70.0	2.00	5.60
Peer and context	7	25.0	75.0	1.75	5.25
Working score				24.00	78.85

The working score is confined to 24.0–78.85 of a nominal 0–100 range. Simulation over 200,000 parcel-seasons with plausible feature distributions gives a mean of 45.4 and a standard deviation of 5.17, with an observed range of 27.1–65.0. The distribution is approximately symmetric but flatter-topped than Gaussian, and it is not normal: a Shapiro–Wilk test rejects at $p \approx 2 \times 10^{-4}$.

10.2 Why absolute cutpoints fail on this distribution

Applying the fixed cutpoints A 0–20, B 21–40, C 41–60, D 61–80, E 81–100 to that distribution produces the following occupancy:

Band	Share of parcels	Note
A	0.0%	Unreachable — the working score cannot fall below 24.0.
B	15.2%	
C	79.6%	Four parcels in five.
D	0.1%	
E	0.0%	Unreachable — the working score cannot exceed 78.85.
(gap)	5.2%	Falls between integer cutpoints; mislabelled A by the implementation.

> *A rating scale that places four parcels in five into one grade provides no discrimination, and discrimination is the only purpose a rating scale serves. A supervisory review of a rating system with this concentration would not accept it. This is a construction property, not a data artefact: it occurs for any plausible feature distribution.*

Two design faults compound it. The bands are roughly four standard deviations wide, so they cannot separate a population with a standard deviation near five. And two components — soil and land, and management quality — are scored from the same feature and correlate at $r = +0.999$, so the apparent seven-dimensional model carries closer to five dimensions of information.

10.3 The banding rule

The published index value is the percentile of the working score within a multi-season reference distribution for the parcel's cohort. Bands are then fixed percentile ranges.

$$\text{published score} = \text{percentile of working score within the cohort reference distribution}$$

Band	Percentile range	Meaning
A	$0 \le p < 20$	Ranks better than at least 80% of comparable parcel-seasons.
B	$20 \le p < 40$	
C	$40 \le p < 60$	Mid-range against the reference.
D	$60 \le p < 80$	
E	$80 \le p \le 100$	Ranks in the worst fifth of comparable parcel-seasons.

Element	Specification
Cohort	Crop × governorate. Where a cohort holds fewer than 30 parcel-seasons, widen to crop × region, then to crop nationally; the cohort actually used is recorded with the value.
Reference distribution	Pooled working scores for the cohort across at least five prior seasons, spanning good and poor years. Rebuilt annually, versioned, and retained so historical values remain reproducible.
Why multi-season, not within-season	Within-season ranking would force exactly 20% into band E every year, including in an excellent season. Anchoring to a multi-season reference lets the whole book move: grades remain comparable across seasons, which is what provisioning and rating-grade mapping require.
Band intervals	Half-open, so no value falls between bands. This removes the cutpoint-gap defect by construction.

10.4 Behaviour across seasons

Simulated band occupancy under the rule above, for seasons of differing quality against a fixed eight-season reference:

Season	A	B	C	D	E
Very good	37.5%	24.6%	17.7%	13.2%	7.0%
Good	28.3%	23.5%	19.9%	16.7%	11.6%
Neutral	19.7%	21.2%	21.6%	20.4%	17.1%
Poor	10.0%	15.9%	20.1%	24.6%	29.4%
Drought	4.6%	9.5%	15.3%	24.5%	46.2%

Every band is reachable, every band is populated in a normal season, and a drought is visible as a shift of the whole book toward D and E rather than as a scale that saturates.

10.5 The FRI requires its own reference distribution

A farm score is an area-weighted mean of parcel scores, and averaging compresses dispersion — in simulation the farm-level standard deviation is roughly 56% of the parcel-level figure. Percentiling a farm score against the parcel reference distribution therefore reintroduces exactly the concentration the rule was designed to remove, placing about two thirds of farms in band C.

The FRI must be percentiled against a reference distribution built from farm scores, constructed the same way and versioned alongside it. With that correction, farm bands populate and respond to season quality in the same manner as parcel bands.

> *This is the single most common implementation error in the redesign: inheriting the parcel band scale for farms. The two reference distributions are different objects and must be stored, versioned and retrieved separately.*

10.6 Reading a band

A band remains a population statement. Band D means the parcel ranks in the fourth quintile of comparable parcel-seasons; it does not predict that this parcel will fall short. The percentile formulation makes this easier to state honestly than the absolute score did, because the number now carries its comparison set explicitly.

11. What this means for a financial institution

11.1 The mapping a bank must perform

Zr3i supplies a production-risk driver. The bank maps the band to its own probability of default and owns that mapping. This is not a commercial preference; under an internal-ratings-based approach a bank cannot outsource its PD estimation to a vendor, and any vendor claiming to supply PD directly is either misdescribing its output or inviting a supervisory problem.

Zr3i supplies	The bank derives
CRI and FRI, 0–100, with band	Its own PD, LGD and EAD
Observed shortfall frequency per band, once calibrated	The rating-grade mapping
Component decomposition and feature values	Explanation for the credit file
Method version and accuracy register reference	Model risk documentation

11.2 Where the index adds value in the credit process

Stage	Use	Constraint
Origination	Rank a pipeline; identify which parcels warrant a site visit.	Not a decision rule. A band cannot approve or decline.
Structuring	Size exposure against concentration — the FRI adjustments show where a farm is fragile.	Advisory. Limits remain the bank's.
Monitoring	Detect in-season divergence from the parcel's own norm before harvest.	Two independent passes required before any alert.
Provisioning	Evidence of condition change, as supporting documentation.	Does not substitute for the bank's impairment model.
Portfolio review	Concentration analysis across canal commands and crops.	Descriptive, not predictive of correlated default.

11.3 What a validation team should require before use

- Completed calibration with the acceptance test in section 13 passed out of sample.
- The accuracy register for the relevant crop and governorate strata, including strata that failed.
- Evidence that the band boundary defect (Annex A, Gap A-2) has been remediated.
- Confirmation of which components were established for each scored parcel, since renormalisation changes effective weights.
- The method version identifier recorded against every value used in a credit decision.

12. What this means for an insurance company

13.1 Underwriting and pricing

For an insurer the index is a risk-classification input, not a rate. It supports differentiation within a book — distinguishing parcels whose conditions differ materially — and it supports exclusion decisions at the extremes. It does not produce a pure premium, because it is calibrated to a shortfall threshold rather than to a loss distribution.

Application	How the index is used	Limit
Risk classification	Segment a portfolio by band before rating.	Bands are relative, not absolute loss costs.
Pre-bind verification	Confirm the parcel exists, its area, and its condition at inception.	Verification of fact, not of intent.
Basis risk reduction	Parcel-level observation narrows the gap between an area index and the individual farmer's loss.	Basis risk is reduced, not eliminated.
Loss adjustment support	Independent condition evidence through the season, timestamped and version-bound.	Evidence for adjusters, not an adjudication.
Portfolio accumulation	Spatial concentration across canal commands and governorates.	Correlated-loss modelling remains the insurer's.

13.2 Basis risk — the honest position

Index insurance fails when the index moves and the farmer's loss does not, or the reverse. Parcel-level optical observation materially narrows that gap relative to an area-yield or weather-station index, and the literature supports this. But the index is a condition measure, not a loss measure: it cannot see a localised pest outbreak that the canopy masks, nor a harvest-time market failure. An insurer using CRI as a trigger rather than as a classification input inherits that residual basis risk directly.

> *The accuracy register is the artefact that makes basis risk quantifiable rather than assumed. It reports measured agreement per crop, governorate and size stratum — including strata where agreement is too low to publish. An insurer should require it before relying on the index at all.*

12.3 Why the CRI cannot be a claim trigger

The CRI is a risk measure. It blends water security, climate exposure, soil, yield stability, management, tenure documentation and peer ranking into a single number. Three of those — tenure, peer context and management — carry no information about whether a loss occurred in a given season. Triggering an indemnity on the CRI would therefore pay out partly on the quality of a grower's land documentation and partly on how neighbouring parcels happened to perform.

> *A risk index and a loss index are different objects with different validation obligations. The CRI classifies exposure before the event; a loss index estimates what happened after it. Using the first as the second is a category error, and a reinsurer will identify it immediately.*

Claims therefore require a second, single-purpose index. The remainder of this section specifies it.

12.4 The Loss Index — specification

Attribute	Specification
Purpose	Estimate realised yield for a parcel-season, in tonnes per feddan.
Inputs	The seasonal vegetation integral alone — cumulative NDVI across the crop cycle, masked and aggregated as in section 5. No weighted components, no tenure, no peer ranking, no management term.
Calibration	Segmented regression of realised yield on the log-transformed integral, fitted per crop and governorate cohort against harvest records.
Output	An estimated yield with its calibration error term, not a score and not a band.
Relationship to the CRI	Independent. The two share observation and feature layers but nothing above them. Neither is an input to the other.

12.5 Payout arithmetic

Two contract forms are supported. Both are standard in index insurance and both are computed by the insurer, not by Zr3i.

$$\text{reference yield } y_r = \text{parcel's own five-season median}$$

$$\text{trigger } T = \text{coverage} \times y_r \qquad \text{exit } X = \text{exit level} \times y_r$$

Indemnity form

$$\text{indemnity} = \max(0,\ T - \hat{y}) \times \text{insured area} \times \text{price}$$

Parametric form

$$\text{payout ratio} = \text{clamp}((T - \hat{y}) / (T - X),\ 0,\ 1)$$

$$\text{indemnity} = \text{payout ratio} \times \text{maximum payable}$$

Where maximum payable is set to the value of the shortfall at exit — (T − X) × area × price — the parametric form reproduces the indemnity form exactly, while remaining simpler to administer and to reinsure. That equivalence is worth preserving when the contract is drafted.

12.6 Worked claim example

Parcel ZR-KS-03, maize, 55.7 feddan. Reference price 8,800 EGP per tonne from price book RPB-2026-01.

Parameter	Value
Reference yield (five-season median)	3.40 t/fd
Coverage level	80%
Trigger T	2.72 t/fd
Exit level	50%
Exit X	1.70 t/fd
Maximum payable (T − X) × area × price	499,963 EGP
Loss Index estimate ŷ	2.10 t/fd
Shortfall (T − ŷ)	0.62 t/fd
Payout ratio	0.6078
Indemnity	303,899 EGP

Both contract forms give 303,899 EGP for this loss, as expected given the maximum payable is set to the shortfall value at exit.

Estimated yield ŷ	Shortfall	Payout ratio	Indemnity EGP
3.20	0.00	0.0000	0
2.90	0.00	0.0000	0
2.72	0.00	0.0000	0
2.50	0.22	0.2157	107,835
2.10	0.62	0.6078	303,899
1.90	0.82	0.8039	401,931
1.70	1.02	1.0000	499,963

Estimated yield ŷ	Shortfall	Payout ratio	Indemnity EGP
1.50	1.22	1.0000	499,963

12.7 The precision required to settle claims

The indemnity is a function of the yield estimate, so the calibration error propagates directly into the amount paid. At the current design-target error for maize of 0.51 t/fd, a one-standard-deviation band around the estimate of 2.10 t/fd runs from 1.59 to 2.61 t/fd — which corresponds to an indemnity anywhere between 53,918 and 499,963 EGP.

Calibration RMSE (t/fd)	Indemnity low	Indemnity high	Precision (± % of central)
0.51 — current design target	53,918	499,963	73%
0.40	107,835	499,963	65%
0.30	156,851	450,947	48%
0.20	205,867	401,931	32%
0.15	230,375	377,423	24%
0.12	—	—	20%
0.10	254,883	352,915	16%

> *At the current design-target accuracy the Loss Index cannot determine an indemnity. A ±73% band on the settled amount is not a basis on which any insurer can pay. Reaching ±20% precision requires an RMSE near 0.12 t/fd — a 76% reduction. Until that is demonstrated, the Loss Index supports loss adjustment; it does not perform it.*

Widening the contract bands does not help. Moving the exit from 50% to 10% of reference yield leaves the percentage precision essentially unchanged, because the error and the payout scale together. The only lever is calibration accuracy.

12.8 Division of responsibility

Zr3i supplies	The insurer owns
Verified area, crop and boundary at inception	The underwriting decision and the rate
Condition time series through the season, version-bound	The policy wording, trigger and exit levels
Two-pass confirmed alerts on material divergence	Whether to instruct an adjuster
Estimated yield with its calibration error term	The indemnity calculation and the settlement
The accuracy register for the relevant stratum	The reserving and reinsurance treatment

Zr3i does not calculate indemnities, does not hold risk, and does not adjudicate claims. The same restraint that keeps the CRI outside a lender's PD estimation applies here: the party that bears the loss owns the determination of it.

13. Validation and calibration protocol

13.1 What is being validated

The claim under test is narrow and falsifiable: that parcels in higher bands fall short of their own yield norm more often than parcels in lower bands, out of sample. Nothing broader is claimed, and nothing broader is testable with the data available before a lending track record exists.

13.2 Protocol

Element	Specification
Panel	40–60 farms across at least four crops and three governorates.
History	Eight seasons, reconstructed from the Sentinel-2 archive, which has been continuous since 2017.
Outcome variable	Yield shortfall — production below 80% of the parcel's own five-season median.
Estimation	Segmented regression on the log-transformed vegetation integral against realised yield.
Resampling	Five-fold cross-validation with 1,000 bootstrap iterations.
Held-out unit	Farms, not seasons. Holding out seasons leaks farm-level information into the test set.
Re-estimation	Component weights re-fitted against observed shortfall; method version increments.

13.3 The acceptance test

One test, specified in advance, determines whether the model may be represented as validated:

> *Band E must show materially higher observed yield-shortfall frequency than band A, out of sample, across at least two crops, with a monotonic curve between them. A flat or non-monotonic curve falsifies the model as specified.*

Stating the falsification condition before the data arrives is the point. A model that cannot be refuted by its own acceptance test is not a model.

14. Governance, versioning and reproducibility

Control	Implementation
Method versioning	Every derived value — observation, feature, component, index, alert — carries a method_id foreign key. Re-running a date under the same method reproduces the identical value.
Specification storage	The method version record stores the full specification as structured data, so a historical score can be re-derived under the ramps in force at the time.
Confidence	Read from a published accuracy register keyed on crop, governorate and size band. Where no register entry exists the system returns "not yet established" rather than a value.
Alerting	Two independent satellite passes are required before any alert is raised. A single pass never fires.
Consent	Every institutional read requires a live, scoped, revocable grant. Refusals are written to an append-only audit log with actor, scope and timestamp.
Immutability	Audit and verification records are append-only, enforced at the database level rather than in application code.

15. Limitations and known exclusions

- The model has not been calibrated against realised outcomes. Per-crop error terms in the specification are design targets.
- Optical observation cannot see below the canopy. Root disease, soil-borne pathogens and sub-canopy pest damage are invisible until they affect canopy state.
- Flooded paddy confounds the optical signal; rice is expected to be the weakest crop in calibration.
- The model has no price, market or counterparty dimension. A farm can score band A and still fail commercially.
- Parcels below roughly five feddan are expected to fall below reliable resolution; the accuracy register is the mechanism for declaring that publicly.
- Crop identity is declared by the operator and is not verified against a field observation; a misdeclared crop is scored against the wrong thresholds and is not detected (Gap A-1).
- Tenure is proxied by area divergence alone. It does not read a land register and makes no legal finding.
- No live satellite ingest has yet been run against production scenes (Annex A, Gap A-5).

16. International standards and frameworks alignment

> *This section identifies established international frameworks the model's construction is consistent with, and states where formal certification or audit against each has and has not been performed. It is a map of alignment, not a claim of compliance. No accreditation body has reviewed this model against any framework listed below.*

16.1 Risk management framework

Framework	Relevance to this model	Alignment status — not certification
ISO 31000:2018 — Risk management: Guidelines	Provides the generic structure this specification already follows: explicit scope and criteria (sections 1–2), a documented basis for each risk weighting (section 7), monitoring and review (sections 4, 13–14), and recorded, versioned decisions (section 14).	Structurally consistent. No third-party conformity assessment performed. ISO 31000 is a guidance standard and is not certifiable — there is no ISO 31000 certificate to hold.

16.2 Financial-sector boundary — Basel Framework

The Basel Framework governs how banks using an internal-ratings-based approach may estimate probability of default, loss given default and exposure at default. This model does not attempt to satisfy Basel IRB requirements directly, and section 11.1 states the reason: a vendor score built on agronomic outcomes cannot substitute for a bank's own PD estimation without misdescribing what it is.

Basel IRB requirement (illustrative)	How this model relates to it — no supervisory review performed
A rating system must discriminate risk meaningfully across grades.	Directly relevant to Gaps A-2 and A-3. The current absolute-cutpoint banding fails this test outright — 79.6% of parcels in one band. The percentile banding rule in section 10.3 is designed to satisfy it once implemented.
Ratings and their inputs must be reproducible and version-controlled.	Satisfied by design through the method-versioning control in section 14.
The institution using the model, not the vendor, owns the PD mapping and its validation.	Stated explicitly in section 11.1. The model is designed to sit below that boundary, not to cross it.
Model risk must be documented, including known limitations.	Sections 1.2, 15 and Annex A serve this function.

A bank relying on this index within an IRB framework remains responsible for demonstrating to its own supervisor that its use of a third-party production-risk score is appropriately validated and documented. This document supports that exercise; it does not complete it.

16.3 Insurance-sector framework — IAIS Insurance Core Principles

The IAIS Insurance Core Principles set supervisory expectations for insurer risk management and for the use of models and data. The principles most relevant to this model are ICP 16, Enterprise Risk Management for Solvency Purposes — which covers risk identification and measurement and underwriting policy — and ICP 14, Valuation, which governs reserving. ICP 17 concerns capital adequacy and is a downstream consequence rather than the direct locus of model governance.

Section 12's treatment of the index as a risk-classification input rather than a rate, its explicit refusal to serve as a claim trigger, and its basis-risk analysis are consistent with the ICP expectation that an insurer using external data or models understands and documents their limitations rather than treating them as authoritative outputs.

No IAIS-affiliated supervisor has reviewed this model. Alignment here means the document's own caveats already match what a supervisor applying the ICPs would look for — not that a supervisory review has taken place. ICP conformity is assessed at the insurer and jurisdiction level, never against a vendor.

16.4 Geospatial data standards

Standard	Relevance	Status — no conformance asserted
ISO 19115-1:2014 — Geographic information: Metadata	Governs the structure of metadata describing geospatial datasets. The parcel boundary, projection and area-computation method in section 5.4 are the kind of provenance ISO 19115 metadata carries.	Not currently published in this format. Doing so would let a bank or insurer's own GIS ingest Zr3i parcel data without bespoke translation.
ISO 19131:2007 — Geographic information: Data product specifications	A template for specifying a geospatial data product — scope, data model, quality requirements, delivery format.	Sections 3–6 already contain most of the substantive content. Formal conformance would mean restating it under ISO 19131 headings.
CEOS Analysis Ready Data (CARD4L)	Community-agreed standards for what counts as analysis-ready optical satellite data.	The Sentinel-2 L2A processing, masking and quality flagging in section 5.1 follow the same principles CARD4L codifies. Formal CARD4L conformance is not asserted.

16.5 Agricultural index insurance literature

There is no single binding international standard for satellite-based agricultural index insurance; the field is guided by peer-reviewed literature and multilateral practice notes rather than an ISO-numbered standard. The two references carried in Annex B — the NHESS systematic review of satellite-based index insurance data, and the study on vegetation-health indices for drought insurance — represent the closest thing to a field consensus, and the basis-risk treatment in sections 12.2 and 12.7 is consistent with both.

16.6 What formal certification would require

None of the above substitutes for certification or supervisory review. If Zr3i intends to make a compliance claim rather than an alignment statement, each item below is a separately scoped undertaking, not a documentation exercise.

- ISO 31000 has no certificate. The closest formal step is an internal-audit or third-party risk-management maturity assessment against it.
- Basel IRB conformity is assessed by a bank's own prudential supervisor, not by Zr3i. Zr3i's role is limited to supplying documentation the bank can use in its own submission.
- IAIS ICP conformity is assessed at the insurer and jurisdiction level by the relevant supervisor, never against a vendor directly.
- ISO 19115 and ISO 19131 conformance would require restructuring the metadata and product specification under those standards' headings, and independent review.
- CARD4L conformance would require formal self-assessment against the CEOS threshold and target requirements, published for peer review.

The honest position in a buyer conversation is that the model is built to be consistent with these frameworks and that none of them has certified it. Claiming more converts a defensible statement into one that fails on the first question.

Annex A — Specification versus implementation

This annex records every known divergence between the specified model above and the running implementation, with evidence. It exists because a validation team cannot assess a model from its specification alone, and because presenting the specification as though it were the implementation would be misleading.

Severity is assessed as the effect on a risk decision, not on software quality.

A-1 — Crop identity is declared, never verified

Severity	Critical
What diverges	The crop attached to a parcel is read from the parcel profile as a declared value. No ground-truth observation, and no classification confidence threshold, is reconciled against it before the parcel is scored.
Evidence	FriEngine reads crop_label from fri_parcel_profile; src/Compute contains no reference to ground_truth or to any crop-verification routine.
Effect on a decision	Crop selects the heat-stress threshold, the phenological window that heat is evaluated in, the reference price book and the per-crop error term. A parcel recorded as wheat but planted with rice is scored against the wrong threshold in the wrong window — climate exposure alone carries 20% of the index — and every monetary figure is wrong. Nothing in the system detects it.
Remediation	Require that a parcel-season may be scored only where crop identity is field-verified, or classified above the published agreement threshold for its stratum. Where neither holds, treat the affected components as not established and renormalise, as the model already does for missing data.
Recommended sequence	Sequence first, before any calibration or external release: it corrupts every downstream feature that depends on crop-specific thresholds, so calibrating before fixing this would fit the model to mislabelled inputs.
Acceptance criterion	For a sample of scored parcel-seasons drawn per the Section 4.2 sampling design, the declared crop matches the field-verified or classifier-confirmed crop at or above the stratum's published agreement threshold; parcels below threshold are excluded from scoring rather than scored against the wrong crop.

A-2 — Index is compressed; two of five bands are unreachable

Severity	Critical
What diverges	The working score is confined to 24.0–78.85 by the component ramps. Bands A and E cannot be reached. Applying the published absolute cutpoints places roughly 80% of parcels in band C and 97% of farms in band C.
Evidence	Derived analytically from the component floors and ceilings in the method specification, and confirmed by simulation over 200,000 parcel-seasons (section 10.1–10.2).
Effect on a decision	The rating scale does not discriminate. A lender grading a book receives one grade for four parcels in five, which is no information. This is the defect that makes the index commercially unusable, independent of calibration.
Remediation	Publish the cohort percentile rather than the working score, and band on fixed percentile ranges against a multi-season reference distribution, as specified in section 10.3. Build a separate farm-level reference for the FRI (section 10.5).
Recommended sequence	Sequence together with A-3, immediately after A-1: the percentile-based banding rule in Section 10.3 resolves both A-2 and A-3 in a single implementation change, and should be

	built and tested before any score is shown to a bank or insurer.
Acceptance criterion	Simulated or production band occupancy across a representative multi-season reference shows every band populated at a rate consistent with Section 10.4's illustrative table (no band at 0% or above roughly 40% under a neutral season), for both the parcel and the separately-built farm reference distribution.

A-3 — Band boundaries leave gaps, and out-of-range scores default to band A

Severity	Critical
What diverges	Band cutpoints are integer-inclusive (A 0–20, B 21–40, C 41–60, D 61–80, E 81–100) while the index is computed to one decimal place. Any score between the integers — for example 40.4, 60.5 — matches no band. The fallback returns band A unless the score exceeds 80.
Evidence	FriSpec::bandFromScore; verified by replaying the published cutpoints against scores 20.5, 40.5 and 60.5, each of which returns A.
Effect on a decision	A parcel scoring 60.5 — upper-moderate risk — is labelled band A, the lowest-risk band. The error is silent and always optimistic. The worked example in section 8.2 (FRI 40.4) triggers it.
Remediation	Make cutpoints continuous (A < 20.5, B < 40.5, and so on) or make the fallback raise an exception rather than return a band.
Recommended sequence	Resolved in the same change as A-2 (Section 10.3's half-open percentile bands eliminate the gap by construction).
Acceptance criterion	Replaying every score in {0.0, 0.1, …, 100.0} against the banding function returns exactly one band for each value, with no exception path silently defaulting to A; a unit test enforcing this is added to the build.

A-4 — Two components are driven by the same feature

Severity	High
What diverges	Soil and land (weight 15) and Management quality (weight 10) both score from ndvi_integral, using different ramps. Twenty-five per cent of the index weight rests on one measurement.
Evidence	componentScorers block of the method specification: both entries name feature ndvi_integral.
Effect on a decision	The two components cannot diverge. A validation team will read this as double-counting a single signal, and the apparent seven-dimensional decomposition overstates the information actually in the model.
Remediation	Give soil and land an independent feature — within-parcel spatial variability of NDVI at peak, or a soil property from survey — and reserve ndvi_integral for management.
Recommended sequence	Sequence after A-1 through A-3 (banding and crop identity are decision-blocking; this is an information-quality issue that affects interpretability, not correctness of any single score). Should be resolved before, not after, the first calibration run, since calibration will otherwise fit two components as if they were independent evidence.
Acceptance criterion	The pairwise correlation between the soil-and-land score and the management-quality score, measured across a representative multi-season sample, is materially below the current r = +0.999 once soil and land is re-derived from an independent feature.

A-5 — Farm-level adjustments are stored, not derived

Severity	High
What diverges	The five FRI adjustments are read from a JSON column on the farm profile and summed. The crop Herfindahl index is recomputed from parcel areas, but the adjustment value attached to it is not derived from it.
Evidence	FriEngine farm loop: adjustment steps are decoded from fri_farm_profile.fri_adjustments and summed without recomputation.
Effect on a decision	Farm-level risk is asserted rather than measured. Two farms with identical structure can carry different adjustments, and nothing in the system detects it.
Remediation	Derive each adjustment from its underlying quantity — HHI, distinct canal commands, distinct irrigation sources, seasons on record, share of area in the worst band — with the mapping published.
Recommended sequence	Can proceed in parallel with A-4; does not block A-1–A-3 remediation but must be closed before the FRI worked example (Section 9.2) is used in any external-facing material, since the adjustment figures there are currently unverifiable as stated.
Acceptance criterion	For any two farms with matching HHI, canal-command count, irrigation-source count, record length and worst-band area share, the computed adjustments are identical; a reconciliation test comparing stored versus recomputed adjustments passes for 100% of farms in the portfolio.

A-6 — No live satellite ingest has been run

Severity	High
What diverges	Copernicus credentials are unset, so the evaluator resolver falls through to the fixture implementation. Every index value in the running system derives from synthetic observations.
Evidence	Absence of a populated copernicus configuration block; EvaluatorResolver returns the fixture evaluator when the live adapter reports unavailable.
Effect on a decision	No figure currently produced by the platform is a measurement. The architecture for live ingest exists and is correct, but it has not been exercised against production scenes.
Remediation	Configure credentials and run one parcel-season end to end against a real scene before any external reliance.
Recommended sequence	This gates every other gap's real-world verification: A-1 through A-5 can be built and unit-tested against fixtures, but none of their acceptance criteria can be confirmed against real conditions until live ingest is running. Recommended as the first infrastructure task, run in parallel with A-1's engineering work.
Acceptance criterion	At least one full parcel-season, for a parcel with independently known ground conditions, is scored end-to-end against a live Sentinel-2 scene (not the fixture evaluator), with the resulting feature and component values reconciled against the manually-verified expected values.

A-7 — The specification and the code disagree on yield stability

Severity	Medium
What diverges	The specification defines a ramp on yield_shortfall_frequency (0→1 mapping to 20→75). The

	implementation instead uses a composite of frequency, depth and coefficient of variation (section 6.2) and prefers it over the specified ramp.
Evidence	FriEngine component loop prefers computeYieldStabilityScore over the feature-derived score for this component only.
Effect on a decision	The documented method and the executed method differ for a component carrying 15% weight. The implemented formula is the better of the two; the specification is the one that is out of date.
Remediation	Update the specification to the composite formula and increment the method version, so the published specification matches what runs.
Recommended sequence	Low engineering cost, no blocking dependency; can be closed immediately as a documentation-and-version-bump task alongside any other change.
Acceptance criterion	The method version is incremented, the specification (this document, Section 7.2) reflects the composite formula as the sole formula — already done in this revision — and no scoring path in the codebase falls back to the simple ramp.

A-8 — Per-crop error terms are declared constants

Severity	Medium
What diverges	The calibration block of the specification carries RMSE values per crop. These are design targets entered by hand, not values fitted to observed yields.
Evidence	calibration.crops block of the method specification; no fitting procedure writes to it.
Effect on a decision	Any accuracy statement derived from these figures would misrepresent design intent as empirical result.
Remediation	Populate only from the calibration run, and mark the field as unpopulated until then.
Recommended sequence	Cannot be closed before Section 13's protocol is executed. Sequenced last by necessity, not by choice.
Acceptance criterion	The calibration.crops block is either empty/marked “not yet calibrated” or populated exclusively by output from the Section 13 calibration protocol, with the acceptance test (Section 13.3) passed; no hand-entered value is present in the field a validation team can inspect.

A-9 — Area is held in two places and they disagree

Severity	Medium
What diverges	The parcel register holds a measured area derived from the boundary geometry. The index module holds an independent area on the parcel profile, used for the area weighting and for all monetary valuation.
Evidence	parcel.area_measured versus fri_parcel_profile.area_feddan; observed to differ by a factor of four for at least one demonstration parcel.
Effect on a decision	The FRI area weighting and every monetary figure rest on an area that may not match the land record, without reconciliation.
Remediation	Derive the index area from the register, or add a reconciliation test that fails the build when the two diverge beyond tolerance.
Recommended	Straightforward data-integrity fix; recommended early, alongside A-5, since both are

sequence	reconciliation problems of the same kind.
Acceptance criterion	For every parcel in the portfolio, parcel.area_measured and fri_parcel_profile.area_feddan are either the same field (preferred) or reconciled within a published tolerance; a build-time or nightly test flags any divergence beyond that tolerance.

A-10 — Confidence is not bound to the index value

Severity	Medium
What diverges	Confidence is read from the published accuracy register and surfaced as a separate screen attribute. It does not travel with the CRI or FRI value through the API or the report.
Evidence	ConfidenceReader is invoked by screen controllers; no confidence reference is persisted alongside risk_index or farm_risk_index rows.
Effect on a decision	A score can be lifted from a report or an API response and used without its stratum reference or its measured agreement. A validation team cannot reconstruct, after the fact, how much weight a given value should have carried.
Remediation	Persist the accuracy register reference and the measured agreement with every index row, and return both in the API payload alongside the score.
Recommended sequence	Can be implemented independently of the other gaps; recommended before any API payload is relied on by an external system, since it is the mechanism that makes every other gap's status auditable per value.
Acceptance criterion	Every index value returned by the API or shown in a report carries its accuracy-register reference and measured agreement figure as fields on the same record, not as a separately-queried attribute; a schema check confirms the fields are non-null wherever the register entry exists.

A.11 Summary

Gap	Severity	Blocks external reliance?
A-1 Crop identity declared, never verified	Critical	Yes
A-2 Index compressed; bands A and E unreachable	Critical	Yes
A-3 Band boundary gaps default to A	Critical	Yes
A-4 Soil and management share one feature	High	Yes
A-5 Farm adjustments stored, not derived	High	Yes
A-6 No live satellite ingest	High	Yes
A-7 Specification and code disagree on yield stability	Medium	No — but must be documented
A-8 Error terms are declared constants	Medium	Yes, for any accuracy claim
A-9 Two conflicting areas per parcel	Medium	Yes, for monetary output
A-10 Confidence not bound to the index value	Medium	Yes, for audit

Three gaps change a risk decision rather than a presentation. A-1 corrupts the inputs. A-2 makes the output undiscriminating, and is the one that determines whether the product is usable at all. A-3 mislabels the output. None requires re-architecting the model; A-2 and A-3 are resolved together by the banding rule in section 10.3.

Annex B — References

Sources consulted for the component weights. Each was retrieved and read; none is cited from recollection.

Supports	Reference	Locator
Water	Steduto, P., Hsiao, T.C., Fereres, E. & Raes, D. (2012). Crop Yield Response to Water. FAO Irrigation and Drainage Paper 66. FAO, Rome.	https://www.fao.org/4/i2800e/i2800e00.htm
Water	Yield Responses to Total Water Input from Irrigation and Rainfall in Six Wheat Cultivars Under Different Climatic Zones in Egypt. Agronomy 14(12):3057 (2024).	https://doi.org/10.3390/agronomy14123057
Climate	Ray, D.K., Gerber, J.S., MacDonald, G.K. & West, P.C. (2015). Climate variation explains a third of global crop yield variability. Nature Communications 6:5989.	https://www.nature.com/articles/ncomms6989
Climate	Yield penalty of maize under heat stress in different growth stages: a review. The Crop Journal (2022).	https://www.sciencedirect.com/science/article/pii/S2095311922000223
Climate	High temperature sensitivity of kernel formation in different short periods around silking in maize. Field Crops Research (2021).	https://www.sciencedirect.com/science/article/abs/pii/S0098847220303695
Soil	Influence of small-scale spatial variability of soil properties on yield formation of winter wheat. Plant and Soil (2023).	https://link.springer.com/article/10.1007/s11104-023-06212-2
Soil	Drivers of within-field spatial and temporal variability of crop yield across the US Midwest. Scientific Reports 8:14113 (2018).	https://www.nature.com/articles/s41598-018-32779-3
Management	Yield penalty due to delayed sowing of winter wheat and the mitigatory role of increased seeding rate. European Journal of Agronomy (2020).	https://www.sciencedirect.com/science/article/abs/pii/S1161030120301271
Management	Does shifting from normal to early or late sowing dates provide yield benefits? A global meta-analysis. Field Crops Research (2024).	https://www.sciencedirect.com/science/article/pii/S0378429024003538
Tenure	The impact of land property rights interventions on investment and agricultural productivity in developing countries: a systematic review. Journal of Development Effectiveness 8(1) (2016).	https://www.tandfonline.com/doi/full/10.1080/19439342.2016.1160947
Tenure	Is the relationship between agricultural performance and land tenure security uniform across the world? Evidence from meta-analysis. Frontiers in Sustainable Food Systems (2026).	https://www.frontiersin.org/journals/sustainable-food-systems/articles/10.3389/fsufs.2026.1752236/full
Peer / credit	Do remotely-sensed vegetation health indices explain credit risk in agricultural microfinance? World Development 127 (2020).	https://www.sciencedirect.com/science/article/abs/pii/S0305750X19304206
Insurance	Satellite-based data for agricultural index insurance: a systematic quantitative literature review. NHESS 25:913 (2025).	https://nhess.copernicus.org/articles/25/913/2025/
Insurance	Satellite-based vegetation health indices as a criterion for insuring against drought-related yield losses. Agricultural and Forest Meteorology (2016).	https://www.sciencedirect.com/science/article/abs/pii/S0168192316300016

Annex C — Data sources, licensing and dependency

Every external data source the model depends on, with its provider, licence and the features that fail if it becomes unavailable. A validation team should read the licence column as a statement of continuity risk: a model resting on a discretionary commercial feed carries a different risk profile from one resting on an open public mission.

Source	Provider	What it supplies	Cadence / resolution	Licence	Features dependent on it
Sentinel-2 MSI Level-2A	ESA / European Commission, via the Copernicus Data Space Ecosystem	Atmospherically corrected surface reflectance and the Scene Classification Layer used for masking.	5 days at the equator; 10–20 m	Free and open under the Copernicus data policy; no restriction on commercial use.	ndvi_integral, ndmi_min, ndvi_spatial_cv, ndvi_cohort_percentile, valid_fraction
ERA5-Land reanalysis	ECMWF, via the Copernicus Climate Change Service	Daily maximum and minimum air temperature, used for the heat-stress indicator.	Hourly, aggregated to daily; ~9 km	Free and open under the Copernicus licence for C3S products.	heat_stress
CHIRPS precipitation	Climate Hazards Center, University of California Santa Barbara	Quasi-global precipitation estimates for the rainfall anomaly.	Daily to dekadal; ~5 km	Public domain / open access.	rain_deficit
Copernicus DEM	ESA / European Commission	Elevation and derived slope, used for parcel context and drainage inference.	Static; 30 m	Free and open for non-commercial and commercial use under the Copernicus DEM licence.	Parcel context (not currently a scored feature)
Farm harvest records	The grower or cooperative, captured by Zr3i at onboarding and each season	Realised yield in tonnes per feddan, per parcel per closed season.	Seasonal	Contributed under the consent and entitlement terms; held on the grower's behalf.	yield_shortfall_frequency, worst_shortfall_ratio, reference yield for the Loss Index
Ground-truth field campaign	Zr3i field teams	Crop identity and condition at randomly drawn sample points, with photographic evidence.	Seasonal, November to April	Proprietary to Zr3i.	Accuracy register, crop verification, yield calibration
Parcel boundary and land record	The grower, cooperative or land register	Boundary geometry and declared area.	At onboarding; on change	Contributed under consent; declared values treated as assertions.	area_variance_pct, measured area, area weighting
Reference price book	Published market references, maintained by Zr3i	Crop prices used in valuation and in claim arithmetic.	Seasonal	Compiled by Zr3i from public references; explicitly not a market forecast.	Monetary output, Loss Index indemnity arithmetic

C.1 Continuity and cost

The entire optical and climate stack is free and open, with no commercial licence, no per-scene cost and no usage ceiling. Marginal data cost per additional parcel is therefore effectively zero, and the two sources that do carry cost — the ground-truth campaign and the harvest records — are field operations rather than data purchases.

The corollary is a dependency on public missions continuing. Sentinel-2 has flown continuously since 2015 with a committed successor programme, and ERA5-Land and CHIRPS are established public products. This is a materially lower continuity risk than a commercial imagery contract, and it is one of the reasons the archive-reconstruction approach in section 13 is viable at all.

C.2 What is not used

- No commercial high-resolution imagery. Parcels below roughly five feddan are consequently expected to fall below reliable resolution.
- No proprietary weather station network. Climate inputs are gridded reanalysis, which carries its own spatial representativeness error at parcel scale.
- No third-party credit bureau, land registry API or financial data. The model contains no borrower-level information of any kind.

Annex D — Glossary

Term	Meaning
AWCRI	Area-weighted mean of a farm's parcel CRI values, before farm-level adjustments.
Band	Categorical grade A–E assigned from the index value.
Component	One of seven interpretable risk dimensions, each scored 0–100.
Established	A component for which a score could be computed. Unestablished components are excluded and the weight renormalised.
Feature	A season-level scalar derived from observations; the input to a component.
Feddan	Egyptian unit of area; 4,200.83 m².
Insufficient	A parcel-season whose valid fraction is below 0.60; excluded from the farm roll-up.
Method version	Identifier binding a specification to every value derived under it.
NDMI	Normalised Difference Moisture Index; canopy water content.
NDVI	Normalised Difference Vegetation Index; canopy greenness.
Shortfall	A closed season in which yield fell below 80% of the parcel's own median.
Valid fraction	Share of a parcel's pixels surviving cloud and quality masking.